

The keyboard controller node was c

[illegible]

The Stanley Controller node was i

During operation, the controller first finds the nearest path segment to the turtle's current position. It then computes the lateral or cross-track error, which measures the perpendicular distance from the turtle to the path, and the heading error, which represents the angular difference between the path direction and the turtle's heading. A lookahead point is also used to estimate the curvature of the path ahead, allowing the controller to anticipate turns smoothly.

and stability while tracking the path. The steering angle is constrained within physical limits, and the linear velocity is adaptively reduced during sharper turns to maintain smooth and stable motion.

Finally, the computed linear and angular velocities are published as Twist messages, driving the turtle in real time. The controller also accumulates and prints the total lateral error in the terminal, providing a quantitative measure of tracking performance. The overall implementation successfully achieves path-following behavior with adaptive steering and speed control.

**Image 1:**

```

maximgorkim@maximgorkim: ~ 100x27
Finished <<< turtle_controller [7.64s]

Summary: 1 package finished [8.87s]
maximgorkim@maximgorkim:~$ source install/setup.bash
maximgorkim@maximgorkim:~$ ros2 run turtle_controller stanley_controller
[INFO] [1761050881.380554939] [stanley_controller]: Stanley running (k=1.80, V_max=0.50, V_min=0.00,
k_omega_gain=1.20).
[INFO] [1761050881.380703772] [stanley_controller]: FF active (L_WHEELBASE=0.10, L_LOOKAHEAD=0.40, k
ff=1.00).
[INFO] [1761050882.130568989] [stanley_controller]: Total Lateral Error: 0.0008
[INFO] [1761050884.145915738] [stanley_controller]: Total Lateral Error: 0.0025
[INFO] [1761050886.146511348] [stanley_controller]: Total Lateral Error: 0.0805
[INFO] [1761050888.161821384] [stanley_controller]: Total Lateral Error: 0.0188
[INFO] [1761050890.162043542] [stanley_controller]: Total Lateral Error: 0.0031
[INFO] [1761050892.177905127] [stanley_controller]: Total Lateral Error: 0.0000
[INFO] [1761050894.178093191] [stanley_controller]: Total Lateral Error: 0.0086
[INFO] [1761050896.194028338] [stanley_controller]: Total Lateral Error: 0.0568
[INFO] [1761050898.194288157] [stanley_controller]: Total Lateral Error: 0.0182
[INFO] [1761050900.194459483] [stanley_controller]: Total Lateral Error: 0.0046
[INFO] [1761050902.209665534] [stanley_controller]: Total Lateral Error: 0.0009
[INFO] [1761050904.225923940] [stanley_controller]: Total Lateral Error: 0.0425
[INFO] [1761050906.241958391] [stanley_controller]: Total Lateral Error: 0.0254
[INFO] [1761050908.242167038] [stanley_controller]: Total Lateral Error: 0.0127
[INFO] [1761050910.258070983] [stanley_controller]: Total Lateral Error: 0.0042
[INFO] [1761050912.274168915] [stanley_controller]: Total Lateral Error: 0.0011
[INFO] [1761050914.289906530] [stanley_controller]: Total Lateral Error: 0.0749
  
```

**Image 2:**

